

2D LiDAR Mapping

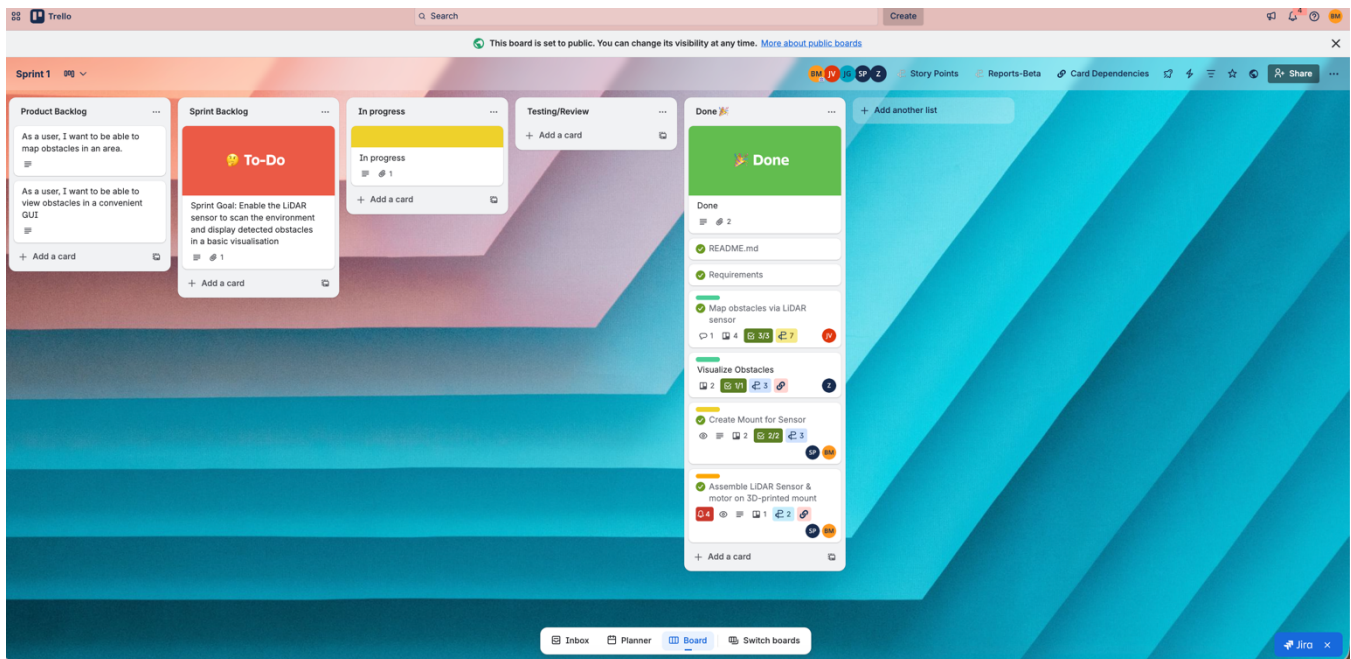
LiStars

Zac Swisher, Sneha Patel, Jamil Velez, Bryton Montgomery

GitHub repository: <https://github.com/swisherrr/2D-lidar-mapping>

Trello board: <https://trello.com/w/listars2/>

Trello board



Done 🏆

🔊 🖨️ ... ✕

✓ Map obstacles via LiDAR sensor

+ Add ⌚ Dates ☑ Checklist 📎 Attachment

Members Labels Story Points Card Dependencies by Screenful

JV +

Must Have +

7

Dependencies

Description

Add a more detailed description...

🔗 Card Dependencies

Remove...

Blocks	Data Export
Blocks	Maps obstacles within radius 360 degrees
Blocks	Maps obstacles within radius 360 degrees
Blocks	Visualize Obstacles

☑ Checklist

Hide checked items

Delete

100%

- ✓ Get lidar sensor working
- ✓ create data pipeline script to retrieve data from sensor
- ✓ save raw data

Add an item

Comments and activity

Hide details

Write a comment...

BM

Bryton Montgomery left this card

[1 hour ago](#)

JV

Jamil Velez moved this card from In progress to Done 🏆

[1 hour ago](#)

Z

zs moved this card from Sprint Backlog to In progress

[Nov 8, 2025, 4:50 PM](#)

JV

Jamil Velez marked this card as complete

[Nov 3, 2025, 11:06 AM](#)

JV

Jamil Velez [Nov 2, 2025, 3:23 PM](#) (edited)

Added the program to Github repo along with requirements.txt.

When LiDAR connected, run the script, and it will scan the area. From the data captured, it will save it onto "scans.csv"

@ • Reply • Delete

JV

Jamil Velez completed save raw data on this card

[Nov 2, 2025, 3:22 PM](#)

JV

Jamil Velez completed create data pipeline script to retrieve data from sensor on this card

[Nov 2, 2025, 3:22 PM](#)

JV

Jamil Velez completed Get lidar sensor working on this card

[Oct 28, 2025, 4:01 PM](#)

Z

zs left this card

[Oct 28, 2025, 3:57 PM](#)

Done 🏆

🔊 🖨️ ⋮ ✕

Visualize Obstacles

+ Add ⌚ Dates ☑ Checklist 📎 Attachment

Members Labels Story Points Card Dependencies by Screenful

Z + **Must Have** + **3** Dependencies

Description

Add a more detailed description...

Card Dependencies

[Remove...](#)

Blocks [Data Export as SVG](#)

Is blocked by [Map obstacles via LiDAR sensor](#)

Checklist

Hide checked items

Delete

100%

- ☒ Create python script that takes in the raw obstacle data and displays obstacles in a GUI

Add an item

Comments and activity

Hide details

Write a comment...

- BM** **Bryton Montgomery** marked this card as complete [1 hour ago](#)
- JV** **Jamil Velez** moved this card from In progress to Done 🏆 [1 hour ago](#)
- JV** **Jamil Velez** completed Create python script that takes in the raw obstacle data and displays obstacles in a GUI on this card [1 hour ago](#)
- Z** **zs** moved this card from Sprint Backlog to In progress [Nov 8, 2025, 4:50 PM](#)
- Z** **zs** removed **Jamil Velez** from this card [Oct 28, 2025, 3:57 PM](#)
- Z** **zs** joined this card [Oct 28, 2025, 3:57 PM](#)
- JV** **Jamil Velez** joined this card [Oct 14, 2025, 4:20 PM](#)
- Z** **zs** attached [Blocks Data Export as SVG](#) to this card [Oct 14, 2025, 3:18 PM](#)
[Reply](#)
- Z** **zs** attached [Is blocked by Map obstacles via LiDAR sensor](#) to this card [Oct 14, 2025, 3:14 PM](#)
[Reply](#)
- Z** **zs** added Checklist to this card [Sep 26, 2025, 3:35 PM](#)

Done 🏠



✓ Create Mount for Sensor

+ Add ⌚ Dates ☑ Checklist 📎 Attachment

Members



Labels



Story Points



Card Dependencies by Screenful

Dependencies

☰ Description

Create Mount for Sensor to attach to wheels

Edit

🔗 Card Dependencies

[Remove...](#)

Blocks Assemble LiDAR Sensor & motor on 3D-printed mount

Blocks Assemble LiDAR Sensor & motor on 3D-printed mount

☑ Checklist

Hide checked items

Delete

100%

✓ Design a mount for Sensor

✓ Print the mount via maker space

Add an item

💬 Comments and activity

Hide details

Write a comment...

BM Bryton Montgomery joined this card
[1 hour ago](#)

JV Jamil Velez moved this card from Sprint Backlog to Done
[1 hour ago](#)

SP Sneha Patel marked this card as complete
[3 hours ago](#)

SP Sneha Patel completed Print the mount via maker space on this card
[3 hours ago](#)

SP Sneha Patel completed Design a mount for Sensor on this card
[3 hours ago](#)

SP Sneha Patel joined this card
[Oct 14, 2025, 3:20 PM](#) 📍

Z zs attached [h](#) Oct 14, 2025, 3:20 PM [nsor & motor on 3D-printed mount](#) to this card
[Oct 14, 2025, 3:15 PM](#)
Reply

Z zs attached [Blocks Assemble LiDAR Sensor & motor on 3D-printed mount](#) to this card
[Oct 14, 2025, 3:14 PM](#)
Reply

Z zs added Checklist to this card
[Oct 14, 2025, 2:59 PM](#)

Done 🏆

🔊 | 🖨️ | 👁️ | ... | ✕

✓ Assemble LiDAR Sensor & motor on 3D-printed mount

+ Add ⌚ Dates ☑ Checklist 📎 Attachment

Members Labels Story Points

SP BM + Could Have + 2

Card Dependencies by Screenful

Dependencies

Description

ie., Sensor rotates 360 degrees

Edit

🔗 Card Dependencies

Remove...

Is blocked by [Create Mount for Sensor](#)

Comments and activity

Hide details

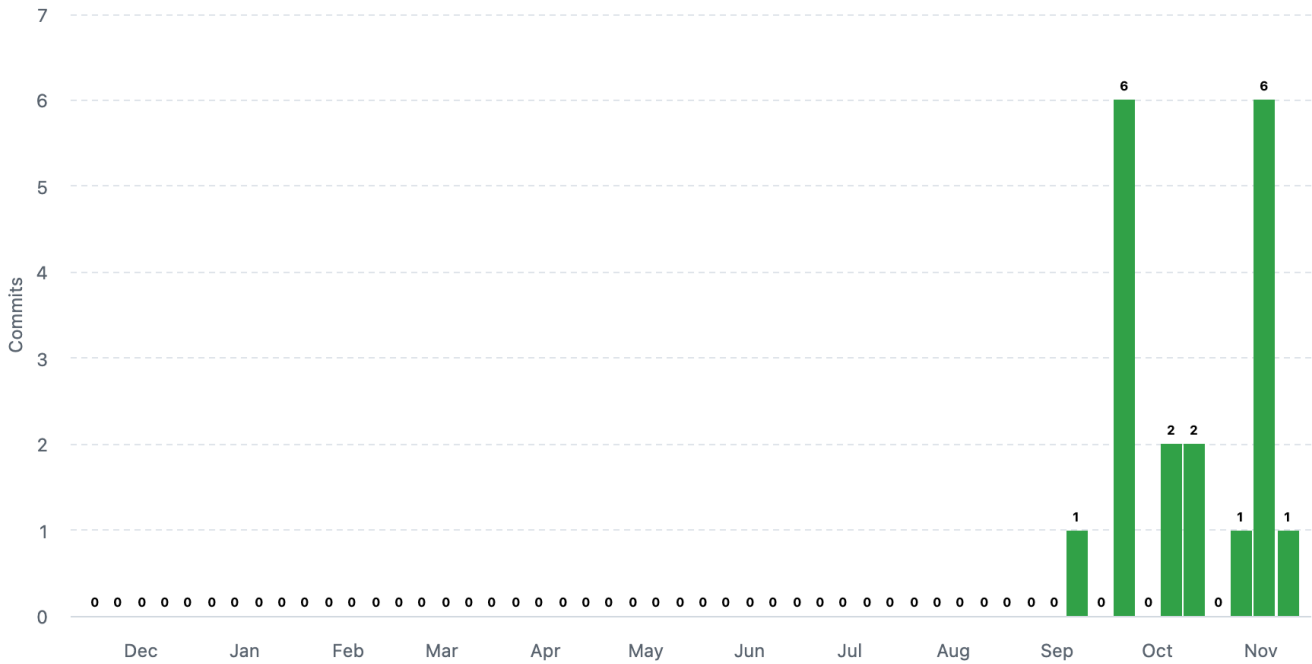
Write a comment...

- JV** **Jamil Velez** moved this card from Sprint Backlog to Done
[2 hours ago](#)
- SP** **Sneha Patel** marked this card as complete
[3 hours ago](#)
- SP** **Sneha Patel** marked this card as incomplete
[3 hours ago](#)
- SP** **Sneha Patel** marked this card as complete
[3 hours ago](#)
- SP** **Sneha Patel** marked this card as incomplete
[3 hours ago](#)
- SP** **Sneha Patel** marked this card as complete
[3 hours ago](#)
- BM** **Bryton Montgomery** joined this card
[Oct 21, 2025, 8:40 AM](#)
- SP** **Sneha Patel** joined this card
[Oct 14, 2025, 3:21 PM](#)
- Z** **zs** attached [Is blocked by Create Mount for Sensor](#) to this card
[Oct 14, 2025, 3:14 PM](#)
[Reply](#)
- Z** **zs** moved this card from Sprint 1 to Backlog
[Sep 26, 2025, 3:14 PM](#)

GitHub contributions

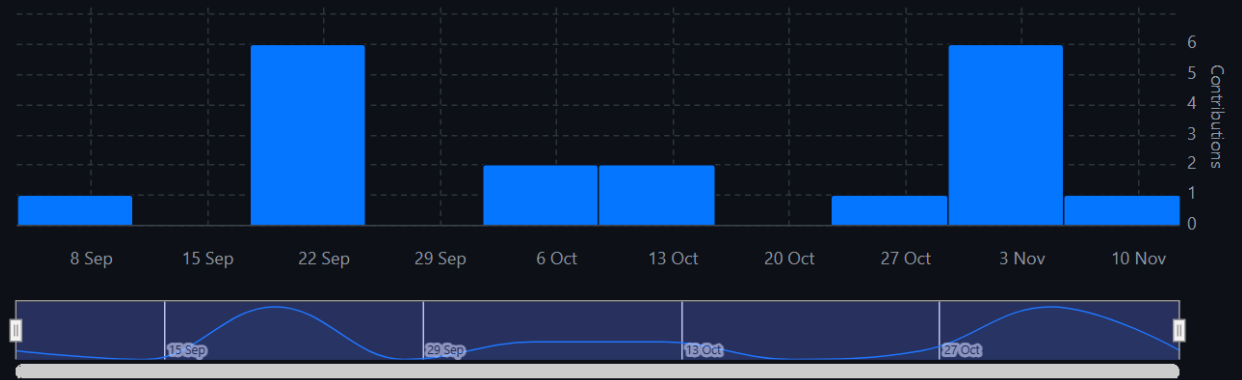
Commits

Number of commits per week



Commits over time

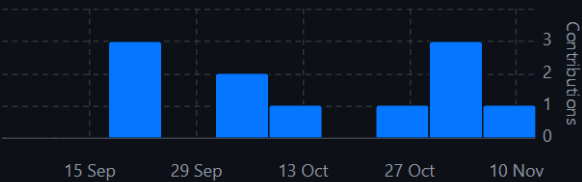
Weekly from Sep 6, 2025 to Nov 8, 2025



JamilVelez

11 commits 577 ++ 82 --

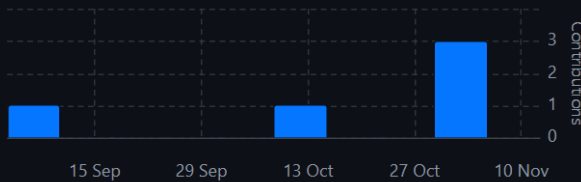
#1 ...



swisherr

5 commits 379 ++ 272 --

#2 ...



sneha0832

2 commits 13 ++ 13 --

#3 ...



Ghostb9

1 commit 2 ++ 2 --

#4 ...

